

PREDICTION TASK  <i>What is the type of task?</i> <i>Which entity are predictions made on?</i> <i>What are the possible outcomes to predict?</i> <i>When are outcomes observed?</i> Type of task: - Supervised learning (regression) Entity: - A residential property, based on 79 parameters. Outcomes: - Final sale price (in USD) When: - After the sale is done, each customer has a contractual obligation to deliver the final price of the asset.	DECISIONS  <i>How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process / application for this.)</i> Predictions are turned into actionable recommendations to the real estate experts of Inved Corp. If the predicted price is significantly lower than the customer's expectations, the consultant uses "Feature Importances" to recommend renovations (e.g., improving KitchenQual). If the price is consistent, the property is listed with a "Price certified by Inved AI" label. Experts interact with the model through an internal web form. The UI instantly returns predicted price, confidence interval, and top 3 features of the property, influencing the price. The model is an assistive tool, not fully autonomous. Experts must manually review the prediction and adjust for unquantifiable factors before final approval.	VALUE PROPOSITION  <i>Who is the end beneficiary, and what specific pain points are addressed?</i> End beneficiaries: - Inved Corp's Real Estate Consultants - Their clients (buyer & sellers) Pain Points addressed: The slow, manual process of traditional property valuation. It eliminates subjective pricing bias, prevents costly human errors (incorrect pricings). How is the ML solution integrating with their workflow? The model acts as the first line of evaluation. Before one of our experts meets a client or physically inspect a property, they will run the property's specifications through the model to get a highly accurate baseline price. The consultant then can use their human expertise to make minor, final adjustments. How will the ML solution integrate with their workflow, and through which user interfaces? It will be integrated into Inved Corp's internal CRM system, as a web dashboard. Consultants will input key property features into a form, and the UI will output the predicted sale price.	DATA COLLECTION  <i>How is the initial set of entities and outcomes sourced?</i> Initial training set is extract from Inved Corp's database, it consists of 1'460 historical real estate transactions. What strategies are in place to update data continuously while controlling cost and maintaining freshness? Continuous Ingestion Through our B2B data integration with Realtor, new data are pulled automatically in real time. Ground Truth Acquisition We continuously update our outcomes by syncing with our internal sales. Whenever a consultant closes a property sale, the data is appended to the property's profile in the database. Freshness By Retraining: To prevent data drift (e.g., changes in the housing market due to inflation or seasons), an automated pipeline triggers model retraining at the end of every month using a rolling window of the most recent data.	DATA SOURCES  <i>Where can we get data on entities and observed outcomes?</i> Internal: Inved Corp's historical SQL database containing past property data and final sale prices. External Partner Source: Direct database-to-database integration with Realtor https://www.realtor.com/. As part of our strategic partnership, Realtor continuously pushes real-time market data on our database, while we push our clients to use their platform.
IMPACT SIMULATION  <i>What are the cost/gain values for (in)correct decisions?</i> Overestimation (false positive): The property remains on the market for too long, leading to decreased customer satisfaction, wasted time for the consultant, and the risk of not receiving commission. Underestimation (false negative): Quick sale but loss of commission for the company and financial loss for the selling customer. Which data is used to simulate pre-deployment impact? The realtor test set will simulate the financial performance of the algorithm	MAKING PREDICTIONS  <i>Are predictions made in batch or in real time?</i> Real-time. The model must provide a price as soon as the Inved Corp consultant enters the parameters into the CRM form. How frequently? On demand, potentially dozens of times a day depending on the number of consultants and client appointments. How much time is available for this (including featurization and decisions)? Less than 5 seconds. The user interface must be fluid so as not to slow down the		BUILDING MODELS  <i>How many models are needed in production?</i> A single regression model. When should they be updated? Once a month, via an automated pipeline using the most recent data, to avoid data drift (inflation, seasons). How much time is available for this (including featurization and analysis)? A few hours. Retraining can be done in batches overnight during the weekend, so there are no strict time constraints.	FEATURES  <i>What representations are used for entities at prediction time?</i> A structured numerical vector containing 79 qualitative and quantitative property features such as location, structural details, condition rating, interior space, utilities, amenities and garage, sale details. What aggregations or transformations are applied to raw data sources? Imputation: Filling missing values (medians for numerical, logical defaults for categorical values). Encoding: Converting text categories into

compared to previous manual estimates.

What are the criteria for deployment?

The model must beat current human accuracy. Technically, the model is deployed if its Root-Mean-Squared-Error (RMSE) on price logs is below a defined threshold. **(Need to be define by us)**

Are there fairness constraints?

Ensure that the model does not systematically undervalue certain geographic areas due to historical biases (avoid virtual redlining).

consultant when dealing with the customer.

Which computational resources are used?

A cloud-hosted API with standard CPU resources, as inference on tabular data is very resource-efficient.

Which computation resources are used?

A Cloud-based CPU instances, auto-scalable (e.g., AWS EC2 or Google Cloud Compute Engine). Because we are working with tabular data and ensemble trees (not Deep Learning images/text), standard, memory-optimized CPUs are sufficient; expensive GPUs are not required.

Goal & Metric : The algorithmic goal is to minimize prediction error on the SalePrice variable. The loss function optimized during training will be the Root-Mean-Squared-Error (RMSE) calculated on the logarithm of prices. Using the logarithm ensures that prediction errors on expensive and cheap houses will penalize the model equally.

numbers

Normalization: Log transformation of highly skewed variables

Aggregation: Aggregating raw spatial features into broader business metrics (e.g. combining various floor areas into a total)

Outlier Handling: Filtering out extreme anomalies (e.g. erroneous data or non-representative market outliers) to prevent model distortion.

MONITORING

Which metrics and KPIs are used to track the ML solution’s impact once deployed, both for end-users and for the business? How often should they be reviewed?

ML metrics:

A/B Testing (Phased Rollout): For the first 3 months, we will A/B test the system by giving the AI tool to Group A (50% of consultants) while Group B (50%) continues manual appraisals, comparing the accuracy and speed of both groups.

Accuracy Tracking: Continuous monitoring of the Root-Mean-Squared-Error (RMSE) on price logs. Once a property officially sells, the final actual price is compared to the AI’s initial prediction.

Drift Monitoring: We monitor the distribution of our model’s predictions relative to real-time market prices on Realtor.com to detect data drift. ML metrics are analyzed monthly, just before automated retraining.

Business KPIs:

Time Saved: Reduce the average time a consultant spends evaluating a property by 50% (from 4 hours to 2 hours) within the first quarter of deployment.

Conversion Rate: Increase the percentage of listed properties that successfully sell within 90 days by 20% over the next year, driven by highly accurate AI pricing that prevents properties from stagnating on the market.

Manual Correction Rate: Track how often



consultants override the AI. The goal is a manual correction rate of less than 10% within the first year, proving that the model is highly trusted by end-users.

Business KPIs are monitored continuously via a weekly dashboard. ML metrics are analyzed monthly just before the model is automatically retrained



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